

Risk register

Operational environment:

The bank is located in a coastal area with low crime rates. Many people and systems handle the bank's data—100 on-premise employees and 20 remote employees. The customer base of the bank includes 2,000 individual accounts and 200 commercial accounts. The bank's services are marketed by a professional sports team and ten local businesses in the community. There are strict financial regulations that require the bank to secure their data and funds, like having enough cash available each day to meet Federal Reserve requirements.

Asset	Risk(s)	Description	Likelihood	Severity	Priority
Funds	Business email compromise	<i>An employee is tricked into sharing confidential information.</i>	2	2	4
	Compromised user database	<i>Customer data is poorly encrypted.</i>	2	3	6
	Financial records leak	<i>A database server of backed up data is publicly accessible.</i>	3	3	9
	Theft	<i>The bank's safe is left unlocked.</i>	1	3	3
	Supply chain disruption	<i>Delivery delays due to natural disasters.</i>	1	2	2
Notes	<i>How are security events possible considering the risks the asset faces in its operating environment?</i> <i>Security events are possible because doing business with other companies introduces more ways for data to be exposed or compromised. Even though the risk of theft isn't a major concern due to the bank's low crime location, other external factors can still create opportunities for security incidents to occur.</i>				

Likelihood:

Likelihood scores were based on how likely it is for a security issue to happen. On the risk matrix, 1 means rare, 2 means likely, and 3 means almost certain. For example supply chain disruptions due to natural disasters were rated a 1 because those events are unpredictable but data compromise events were rated a 2 since they're more likely given the different ways they can occur.

Severity:

All the risks were rated above a 2 in severity because incidents like business email compromises and theft can have serious effects. Customers have trust in banks to protect their information, and any failure could lead to financial and legal consequences.

Priority:

The financial records leak received the highest total risk score of 9, meaning it's almost certain to happen and would have a major impact on the bank's operations. This high score makes it the top priority for the security team to address before dealing with lower-risk issues.

Asset: The asset at risk of being harmed, damaged, or stolen.

Risk(s): A potential risk to the organization's information systems and data.

Description: A vulnerability that might lead to a security incident.

Likelihood: Score from 1-3 of the chances of a vulnerability being exploited. A 1 means there's a low likelihood, a 2 means there's a moderate likelihood, and a 3 means there's a high likelihood.

Severity: Score from 1-3 of the potential damage the threat would cause to the business. A 1 means a low severity impact, a 2 is a moderate severity impact, and a 3 is a high severity impact.

Priority: How quickly a risk should be addressed to avoid the potential incident. Use the following formula to calculate the overall score: **Likelihood x Impact Severity = Risk**

Sample risk matrix

		Severity		
Likelihood		Low 1	Moderate 2	Catastrophic 3
	Certain 3	3	6	9
	Likely 2	2	4	6
	Rare 1	1	2	3